Enterprise Banking Operations Management Platform

Complete End-to-End ServiceNow Capstone Architecture & Execution Blueprint

Built on ServiceNow

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1. PROJECT OVERVIEW

Objective

Build a fully working enterprise-grade banking application that demonstrates:

- Customer Service Management (CSM)
- IT Service Management (ITSM)
- Online Banking Portal
- Workflow automation
- Approval management
- Fraud investigation
- Resolution target management
- Customer onboarding
- Notifications & alerts
- Reporting & executive dashboards
- Role-based access control
- Event-driven architecture

This project simulates a real-world US retail banking operations platform.

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2. APPLICATION DETAILS

Property	Value

Application Name	Banking Operations Management
Scope	x_bank_ops
Type	Single Scoped Application
Platform	ServiceNow
Architecture	CSM + ITSM Integrated

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3. HIGH LEVEL ARCHITECTURE

Online Banking Portal
↓
Record Producers / Catalog Items
↓
Customer Operations Tables
↓
Business Rules + Script Includes
↓
Flow Designer Automation
↓
Approvals / Notifications / Resolution Targets
↓
Fraud Investigation
↓
ITSM Incident / Change
↓
Resolution + Reporting
↓
Dashboard + KPI Monitoring

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4. COMPLETE MODULE STRUCTURE

Module	Purpose
Customer Management	Customer onboarding
Consumer Loan Management	Loan processing

Credit Card Dispute Management	Customer disputes
Fraud Investigation	Fraud operations
Customer Verification	Identity verification
User Provisioning	Portal login access
ITSM Integration	Incident/change handling
Resolution Target & KPI	Performance tracking
Reporting	Dashboard & analytics

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5. COMPLETE DATA MODEL

5.1 CUSTOMER PROFILE TABLE

Table

x_bank_customer_profile

Purpose

Stores customer master information.

Extends

Base table

Fields

Field Label	Type
Customer Number	Auto Number

Full Name	String
Date of Birth	Date
Gender	Choice
Phone Number	Phone
Email Address	Email
Street Address	String
City	String
State	Choice
ZIP Code	String
SSN	Masked String
Driver License Number	String
Customer Verification Status	Choice
Risk Rating	Choice
Customer Status	Choice
User Account	Reference → sys_user

5.2 LOAN APPLICATION TABLE

Table

x_bank_loan_application

Extends

Task

Fields

Field	Type
Loan Number	Auto Number
Customer	Reference
Loan Type	Choice
Loan Amount	Currency
Annual Income	Currency
Estimated Monthly Payment	Currency
Risk Rating	Choice
Approval Status	Choice
Loan Stage	Choice
Assigned To	Reference
Assignment Group	Reference
Resolution Target Status	Choice

5.3 CREDIT CARD DISPUTE TABLE

Table

x_bank_card_dispute

Extends

Task

Fields

Field	Type
Dispute Number	Auto Number
Customer	Reference

Card Number	String
Transaction ID	String
Transaction Amount	Currency
Dispute Type	Choice
Fraud Suspected	True/False
Resolution Status	Choice
Priority	Choice

5.4 FRAUD INVESTIGATION TABLE

Table

x_bank_fraud_investigation

Extends

Task

Fields

Field	Type
Investigation Number	Auto Number
Related Dispute	Reference
Fraud Type	Choice
Severity	Choice
Investigator	Reference
Investigation Status	Choice
Findings	HTML
Resolution Notes	HTML

5.5 CUSTOMER VERIFICATION TABLE

Table

x_bank_document_verification

Fields

Field	Type
Verification Number	Auto Number
Customer	Reference
Document Type	Choice
Attachment	Attachment
Verification Status	Choice
Verified By	Reference
Compliance Notes	HTML

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6. TABLE RELATIONSHIPS

```
Customer
├── Loan Applications
├── Credit Card Disputes
├── Verification Documents
└── sys_user Account

Credit Card Dispute
└── Fraud Investigation

Fraud Investigation
└── Incident
```

```
Loan Application
└── Approval Records
```

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7. COMPLETE ROLE STRUCTURE

Role	Responsibility
x_bank_ops.customer	Portal access
x_bank_ops.loan_processor	Loan handling
x_bank_ops.fraud_analyst	Fraud investigation
x_bank_ops.operations_manager	Approvals
itil	Incident handling
admin	Full access

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8. GROUP STRUCTURE

Group	Purpose
Banking Customers	Portal users
Loan Processing Team	Loan approvals
Fraud Investigation Team	Fraud handling
Banking Operations Team	Backend operations
ITSM Support Team	Incident resolution

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9. COMPLETE PORTAL ARCHITECTURE

Portal Components

Component	Type
Customer Registration	Record Producer
Consumer Loan Request	Record Producer
Credit Card Dispute	Record Producer
Request Status	Widget
My Loans	Widget
My Disputes	Widget
Upload Verification Documents	Catalog Item

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10. CUSTOMER USER PROVISIONING

Flow

Online Registration
↓
Customer Profile Created
↓
sys_user Created
↓
Role Assigned
↓
Group Assigned
↓
Temporary Password Generated
↓
Welcome Notification Triggered
↓
Portal Access Enabled

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11. COMPLETE SCRIPTING ARCHITECTURE

11.1 CLIENT SCRIPTS

Script	Purpose
SSN Validation	Duplicate prevention
Monthly Payment Calculation	Auto calculation
Loan Amount Validation	Business validation
Fraud Visibility	Dynamic form behavior
Verification Check	Submission restriction

11.2 UI POLICIES

Policy	Action
High loan amount	Show approval section
Fraud suspected	Show fraud fields
Verification incomplete	Disable submit

11.3 DATA POLICIES

Policy	Purpose
SSN mandatory	Data integrity
Email validation	Proper format
Phone mandatory	Required field

11.4 BUSINESS RULES

BR Name	Type
Create Portal User	After Insert
Prevent Duplicate Customer	Before Insert
Auto Risk Rating	Async
Create Fraud Investigation	After Update
Auto Incident Creation	After Insert
Loan Approval Validation	Before Update

11.5 SCRIPT INCLUDES

Script Include	Purpose
CustomerUtils	Customer validation
LoanUtils	Payment calculations
FraudEngine	Fraud logic
NotificationUtils	Event handling
UserProvisioningUtils	sys_user creation

11.6 GLIDEAJAX COMPONENTS

Feature	Purpose
SSN Check	Duplicate validation
Payment Calculator	Dynamic calculation
Customer Lookup	Auto-fill data

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12. FLOW DESIGNER ARCHITECTURE

FLOW 1 — CUSTOMER REGISTRATION

Portal Submission
→ Validate Data
→ Create Customer
→ Create sys_user
→ Assign Role
→ Assign Group
→ Generate Password
→ Trigger Welcome Notification

FLOW 2 — LOAN APPROVAL FLOW

Loan Application
→ Risk Evaluation
→ Manager Approval
→ Operations Approval
→ Final Approval
→ Customer Notification

FLOW 3 — CREDIT CARD DISPUTE FLOW

Customer Dispute
→ Dispute Classification
→ Priority Assignment
→ Fraud Check
→ Investigation Trigger
→ Resolution
→ Closure Notification

FLOW 4 — FRAUD INVESTIGATION FLOW

Fraud Detection
→ Investigator Assignment
→ Evidence Collection
→ High Severity Check
→ Incident Creation
→ Escalation
→ Resolution

FLOW 5 — ITSM INCIDENT FLOW

Fraud Investigation
→ Create Incident
→ ITSM Assignment
→ Resolution
→ Update Fraud Record
→ Close Investigation

13. APPROVAL MATRIX

Loan Amount	Approval
< $10,000	Auto Approval
$10,000 – $50,000	Loan Manager
> $50,000	Senior Approval Board
> $250,000	Risk & Compliance Committee

14. COMPLETE EVENT ARCHITECTURE

Event Name	Trigger
customer.created	Customer onboarding
customer.user.created	sys_user created
loan.submitted	Loan request
loan.approved	Approval
dispute.created	Dispute raised
fraud.detected	Fraud escalation
incident.created	ITSM incident

15. NOTIFICATION ARCHITECTURE

Notification	Trigger

Welcome to Online Banking	customer.user.created
Loan Request Submitted	loan.submitted
Loan Approved	loan.approved
Credit Card Dispute Created	dispute.created
Fraud Investigation Alert	fraud.detected
Incident Assigned	incident.created

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16. COMPLETE RESOLUTION TARGET ARCHITECTURE

Loan Resolution Target

Priority	Duration
High	4 Hours
Medium	8 Hours
Low	24 Hours

Credit Card Dispute Resolution Target

Type	Duration
Fraud	2 Hours
Transaction	8 Hours

Card Issue	24 Hours

Fraud Investigation Resolution Target

Severity	Duration
Critical	1 Hour
High	4 Hours
Medium	8 Hours

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17. REPORTING ARCHITECTURE

Reports

Report	Type
Loan Trends	Line
Fraud Distribution	Pie
Resolution Target Breaches	Bar
Pending Approvals	List
Open Incidents	KPI

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18. DASHBOARD COMPONENTS

Dashboard Widget	Purpose
Open Loans	KPI
Fraud Alerts	Indicator

Resolution Target Breaches	KPI
Pending Disputes	List
Approval Queue	Scorecard

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19. ACL ARCHITECTURE

Table	Restriction
Customer	Own records only
Loan	Processor access
Fraud	Fraud team only
Incident	ITIL users

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20. UI ACTIONS

UI Action	Table
Approve Loan	Loan
Reject Loan	Loan
Escalate Fraud	Dispute
Create Incident	Fraud
Close Investigation	Fraud

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21. IMPORT SET ARCHITECTURE

Excel File
→ Data Source

→ Import Set Table
→ Transform Map
→ Customer Table
→ User Provisioning
→ Notification

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22. COMPLETE END-TO-END EXECUTION FLOW

Customer Registration
↓
Customer Record Created
↓
sys_user Created
↓
Portal Access Enabled
↓
Customer Logs Into Online Banking Portal
↓
Loan Request / Credit Card Dispute
↓
Approval / Fraud Evaluation
↓
Fraud Investigation
↓
ITSM Incident Creation
↓
Resolution
↓
Notification
↓
Dashboard Update
↓
Resolution Target Tracking
↓
Closure

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23. COMPLETE BUILD EXECUTION PLAN

PHASE 1 — APPLICATION SETUP

- Create scoped application
- Create roles
- Create groups
- Configure application menus

PHASE 2 — DATA MODEL

- Create tables
- Create fields
- Create relationships
- Configure related lists

PHASE 3 — FORMS & UI

- Configure forms
- Add sections
- Add form layouts
- Configure workspace

PHASE 4 — ONLINE BANKING PORTAL

- Create portal pages
- Create record producers
- Configure widgets
- Configure login access

PHASE 5 — SECURITY

- Create ACLs
- Configure role restrictions
- Configure portal access

PHASE 6 — SCRIPTING

- Client Scripts
- UI Policies
- Data Policies
- Business Rules
- Script Includes
- GlideAjax

PHASE 7 — AUTOMATION

- Create Flow Designer flows
- Configure approvals
- Configure notifications
- Configure events

PHASE 8 — RESOLUTION TARGET & KPI

- Create resolution targets
- Configure breach conditions
- Configure KPIs

PHASE 9 — REPORTING

- Build reports
- Build dashboards
- Configure homepage widgets

PHASE 10 — FINAL DEMO

Final Scenario

Portal Registration
→ Login
→ Consumer Loan Request
→ Approval
→ Fraud Detection
→ Incident Creation
→ Resolution
→ Notification
→ Dashboard Update

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24. FINAL ENTERPRISE COMPONENT COUNT

Component	Approx Count
Tables	5
Roles	5
Groups	5
Client Scripts	8–10
UI Policies	6–8
Data Policies	4–5
Business Rules	8–10
Script Includes	5

Flow Designer Flows	5
Notifications	6
Events	7
Reports	5
Dashboard Widgets	5
Record Producers	3
ACLs	12–15

Banking Operations Management Platform

Client Requirement Gathering Document & Business Scenarios

Built on ServiceNow

This section represents the discussions between:

- Client
- Business Analyst
- Solution Architect
- Product Owner

before development begins.

This teaches students:

- requirement gathering
- user story analysis
- translating business requirements into technical implementation

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1. CLIENT BACKGROUND

Client Type

US Retail & Consumer Banking Institution

Current Problems

The client currently manages:

- consumer loan requests manually
- customer disputes through emails
- fraud investigations in spreadsheets
- customer onboarding in disconnected systems
- customer verification outside the platform

Business Impact

Because of this:

- approvals are delayed
- fraud visibility is poor
- customer experience is inconsistent
- compliance tracking is difficult
- resolution target monitoring is missing
- there is no centralized operations platform

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2. CLIENT EXPECTATION

The client wants a centralized banking operations platform where:

- customers can register online
- customers can apply for consumer loans
- customers can raise credit card disputes
- backend teams can process requests
- compliance teams can verify customer identity

- fraud teams can investigate suspicious activity
- ITSM teams can resolve banking system issues
- managers can monitor KPI and resolution targets
- customers can track request status through the online banking portal

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3. BUSINESS REQUIREMENT AREAS

Module	Requirement Area
Customer Management	Customer onboarding
Consumer Loan Processing	Loan lifecycle
Credit Card Dispute Management	Dispute handling
Fraud Operations	Fraud investigation
ITSM Integration	Incident handling
Online Banking Portal	Customer self-service
Resolution Target Management	Performance tracking
Reporting	Executive dashboards

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4. CUSTOMER REGISTRATION REQUIREMENT

Scenario 1 — Online Customer Registration

Business Requirement

The bank wants customers to register through the online banking portal.

User Story

As a retail banking customer,
I want to register through the online banking portal,
So that I can access digital banking services.

Functional Requirements

Customer should submit:

- Full Name
- Date of Birth
- Phone Number
- Email Address
- SSN
- Driver License Number
- Address Information

System Should

- validate duplicate SSN
- validate email format
- create customer profile
- create portal user account
- assign customer role
- send welcome notification
- enforce password reset on first login

Technical Components

Requirement	ServiceNow Component
Registration Form	Record Producer
SSN Validation	GlideAjax
Customer Creation	Table
User Creation	Business Rule
Welcome Notification	Notification

Scenario 2 — Backend Customer Onboarding

Business Requirement

Operations teams should manually onboard customers from backend forms.

User Story

> As a banking operations user,
> I want to manually create customer profiles,
> So that walk-in customers can be onboarded efficiently.

Functional Requirements

- backend teams can create customers
- portal users created automatically
- group assignment should happen
- notifications should trigger automatically

Scenario 3 — Bulk Customer Upload

Business Requirement

Operations teams should onboard customers using Excel uploads.

User Story

> As an operations manager,
> I want to upload customer records through Excel,
> So that large-scale onboarding becomes faster.

Functional Requirements

- upload Excel file
- transform customer records
- prevent duplicate SSN creation
- auto-create portal users
- send onboarding notifications

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5. ONLINE BANKING PORTAL REQUIREMENT

Scenario 4 — Portal Login

Business Requirement

Customers should log into the online banking portal using temporary credentials.

User Story

> As a banking customer,
> I want secure portal login access,
> So that I can track my banking requests online.

Functional Requirements

- portal-only access
- password reset on first login
- dashboard access
- request tracking

Scenario 5 — Customer Dashboard

Business Requirement

Customers should access all banking requests from a centralized dashboard.

User Story

> As a banking customer,
> I want to see my loan requests and disputes,
> So that I can monitor request progress.

Functional Requirements

Dashboard should show:

- My Loans
- My Disputes
- Pending Verification Documents
- Notifications
- Resolution Target Status

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6. CONSUMER LOAN MANAGEMENT REQUIREMENT

Scenario 6 — Consumer Loan Application

Business Requirement

Customers should apply for consumer loans digitally.

User Story

> As a banking customer,
> I want to apply for a loan online,
> So that I do not need to visit a branch office.

Functional Requirements

Customer should:

- select loan type
- enter annual income
- request loan amount
- upload supporting documents

System Should

- calculate estimated monthly payment
- evaluate risk rating
- trigger approval workflow
- assign resolution target

Scenario 7 — Loan Approval Workflow

Business Requirement

Loan approvals should follow automated approval routing.

User Story

> As a loan manager,
> I want approvals routed automatically based on loan amount,
> So that manual assignment is reduced.

Functional Requirements

Loan Amount	Approval
< $10,000	Auto Approval

$10,000 – $50,000	Loan Manager
> $50,000	Senior Approval Board
> $250,000	Risk & Compliance Committee

Scenario 8 — Loan Rejection

Business Requirement

Loan applications should be rejected if verification or risk checks fail.

User Story

> As a loan processor,
> I want invalid loan requests rejected automatically,
> So that financial risk is minimized.

Functional Requirements

- rejection reason mandatory
- notification should trigger
- audit logs should capture actions

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7. CREDIT CARD DISPUTE MANAGEMENT REQUIREMENT

Scenario 9 — Credit Card Dispute Creation

Business Requirement

Customers should raise disputes for unauthorized or incorrect card transactions.

User Story

> As a credit card customer,
> I want to report unauthorized charges online,
> So that fraud investigations can start quickly.

Functional Requirements

Customer can:

- select disputed transaction
- choose dispute type
- upload supporting evidence

System Should

- assign priority
- create dispute case
- trigger fraud review if required

Scenario 10 — Dispute Classification

Business Requirement

The system should classify disputes automatically.

User Story

> As a fraud analyst,
> I want disputes categorized automatically,
> So that investigations can start faster.

Functional Requirements

- classify by amount
- classify by dispute type
- assign severity

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8. FRAUD INVESTIGATION REQUIREMENT

Scenario 11 — Fraud Detection

Business Requirement

Suspicious transactions should create fraud investigations automatically.

User Story

> As a fraud analyst,
> I want suspicious banking activity detected automatically,
> So that financial loss is minimized.

Functional Requirements

Fraud triggers:

- large transaction amount
- unusual geographic access
- repeated dispute activity
- multiple failed login attempts
- suspicious ACH transactions

Scenario 12 — Fraud Escalation

Business Requirement

Critical fraud investigations should escalate automatically.

User Story

> As a fraud manager,
> I want critical fraud investigations escalated automatically,
> So that urgent security actions happen immediately.

Functional Requirements

- assign fraud investigation team
- trigger notifications
- create incident
- update resolution target

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9. ITSM INTEGRATION REQUIREMENT

Scenario 13 — Incident Creation

Business Requirement

High-severity fraud events should create ITSM incidents automatically.

User Story

> As an IT support engineer,
> I want incidents generated automatically for fraud alerts,
> So that banking systems can be investigated quickly.

Functional Requirements

- auto-create incident
- assign ITSM support group
- sync status updates

Scenario 14 — Change Request Management

Business Requirement

Banking maintenance activities should follow change management controls.

User Story

As an operations manager,
I want banking maintenance activities managed through change requests,
So that service outages are minimized.

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10. CUSTOMER VERIFICATION REQUIREMENT

Scenario 15 — Verification Document Upload

Business Requirement

Customers should upload verification documents securely through the portal.

User Story

As a banking customer,
I want to upload identity verification documents securely,
So that customer verification can happen digitally.

Functional Requirements

- attachment upload
- verification review
- approval/rejection
- status tracking

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11. RESOLUTION TARGET REQUIREMENT

Scenario 16 — Resolution Target Tracking

Business Requirement

All banking requests should follow defined resolution timelines.

User Story

> As an operations manager,
> I want resolution target tracking for all requests,
> So that operational efficiency is maintained.

Functional Requirements

- response targets
- resolution targets
- breach notifications
- escalation handling

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12. NOTIFICATION REQUIREMENT

Scenario 17 — Customer Notifications

Business Requirement

Customers should receive automated banking updates.

User Story

> As a banking customer,
> I want automated notifications,
> So that I always know my request status.

Functional Requirements

Notifications for:

- registration
- approvals
- rejections
- dispute updates
- password setup
- fraud alerts

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13. REPORTING REQUIREMENT

Scenario 18 — Executive Dashboard

Business Requirement

Management teams should monitor banking operations through dashboards.

User Story

As a banking operations manager,
I want KPI dashboards and reports,
So that I can monitor operational performance.

Functional Requirements

Dashboard should show:

- open loans
- fraud alerts
- resolution target breaches
- pending approvals
- incident trends

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14. SECURITY REQUIREMENT

Scenario 19 — Role-Based Access Control

Business Requirement

Users should only access authorized banking information.

User Story

As a system administrator,
I want role-based access control implemented,
So that banking data remains secure.

Functional Requirements

- customers restricted to own records
- backend access controlled
- fraud access limited
- SSN masking enabled
- ACL implementation required

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15. AUDIT REQUIREMENT

Scenario 20 — Audit Tracking

Business Requirement

All critical banking actions should be audited.

User Story

> As a compliance manager,
> I want audit history for customer operations,
> So that regulatory compliance requirements are met.

Functional Requirements

Track:

- approvals
- rejections
- status changes
- comments

- escalations
- compliance updates

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16. FINAL BUSINESS FLOW

Customer Registration
↓
Portal User Provisioning
↓
Online Banking Login
↓
Consumer Loan Request / Credit Card Dispute
↓
Risk & Compliance Validation
↓
Fraud Investigation
↓
ITSM Incident
↓
Resolution
↓
Customer Notification
↓
Executive Dashboard & KPI Tracking

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17. HOW TO EXPLAIN THIS TO STUDENTS

Use this implementation sequence:

Step	Explanation
Requirement	What the client expects
User Story	Business expectation
Technical Mapping	Which ServiceNow component is used
Build Phase	Actual implementation

Demo End-to-end execution